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ECE 308 Assignment 2

3/4/2026

1) Find Transfer Function θl(s)/ θd(s)
    a. Given
        i. P(s) = θl(s)/Vm(s) = b0/[s(s+a)]
        ii. a = 40.323, b = 61.694
        iii. vm(t) = kp(θd(t) – θl(t))– kv d θl/dt
        iv. θl(s) = Vm(s) * P(s)
    b. Calculations
        i. Vm(s) = kp(θd(s) – θl(s)) - kvs θl(s)
        ii. θl(s) = Vm(s) * P(s)
        iii. θl(s) = b0/[s(s+a)] * [kp(θd(s) – θl(s)) - kvs θl(s)]
        iv. θl(s) * [s(s+a)] = b0kpθd(s) – b0θl(kp + kvs)
        v. θl(s) * [s^2 + (a0 + b0kv)s + b0kp = b0kp θd(s)
    c. Result
        i. θl(s)/ θd(s) = b0kp/(s^2 + (a0 + b0kv)s + b0kp

2) Find Transfer Function X(s)/Xd(s)
    a. Given
        i. θd(s) = zKcXd(s) – (sKc + zKc)X(s)
        ii. Pbb = X(s) / θl(s) = Kbb/s^2
    b. Calculations
        i. X(s) = $Kbb/(s$^2) * θl(s)
        ii. θd(s) ~ θl(s)
        iii. X(s) = Kbb/(s^2) * [zKcXd(s) – (sKc + zKc)X(s)]
        iv. (s^2) * X(s) = Kbb* zKcXd(s) – kbb * (sKc + zKc)X(s)
        v. (s^2) * X(s) + kbb * (sKc + zKc)X(s) = Kbb* zKcXd(s)
        vi. X(s) (s^2 + kbb * (sKc + zKc) = Xd(s) * Kbb * zKc
    c. Results
        i. X(s) / Xd(s) = Kbb * zKc / (s^2 + kbb * (sKc + zKc))

3) Find Laplace of Error Signal, Steady State Tracking Error
    a. Given
        i. |ess| ≤ 0.005m, ts = 4s, PO ≤ 10%
        ii. Xd(s) = 1/s
        iii. E(s) = Xd(s) – X(s)
        iv. X(s) / Xd(s) = Kbb * zKc / (s^2 + kbb * (sKc + zKc)
    b. Calculations
        i. X(s) = Xd(s) * Kbb * zKc / (s^2 + kbb * (sKc + zKc)
        ii. E(s) = X(ds)(1 - Kbb * zKc / (s^2 + kbb * (sKc + zKc))
        iii. E(s) = X(ds)[(s^2 + Kbb * Kc * s)/ (s^2 + kbb * (sKc + zKc))]
        iv. E(s) = [(s + Kbb * Kc)/ (s^2 + kbb * (sKc + zKc))]
        v. ess = lim(s -> ∞)[[(s + Kbb * Kc)/ (s^2 + kbb * (sKc + zKc))] = 0
    c. Results
        i. E(s) = [(s + Kbb * Kc)/ (s^2 + kbb * (sKc + zKc))]
        ii. Ess = 0

4) Find values of damping coefficient and undamped natural frequency
    a. Given
        i. |ess| ≤ 0.005m, ts = 4s, PO ≤ 10%
    b. Calculations
        i. 10 = 100 * e^(-zeta * pi / √(1-zeta^2))
        ii. ln(0.1) = -zeta * pi / √(1-zeta^2)
        iii. -ln(0.1)/pi = zeta/√(1-zeta^2)
        iv. ln(0.1)^2/pi^2 = zeta^2/(1-zeta^2)
        v. ln(0.1)^2/pi^2 -zeta^2 * ln(0.1)^2/pi^2 = zeta^2
        vi. ln(0.1)^2 /pi^2 = zeta^2 + zeta^2 * ln(0.1)^2/pi^2
        vii. ln(0.1)^2 /pi^2 = zeta^2 (1 - * ln(0.1)^2/pi^2)
        viii. [ln(0.1)^2 /pi^2] / (1 - * ln(0.1)^2/pi^2) = zeta^2
        ix. Zeta = ln(0.1)/(pi^2 + √(1 - * ln(0.1)^2) = 0.591
        x. Ts = 4 = 4/(zeta * wn)
        xi. Wn = 1/zeta = 1/0.591 = 1.692
    c. Results
        i. Zeta = 0.591
        ii. wn = 1.692

5) Find Gains Kc and z to meet spec
    a. Given
        i. |ess| ≤ 0.005m, ts = 4s, PO ≤ 10%
        ii. kbb = 0.419
    b. Calculations
        i. s^2 + kbb * sKc + kbb * zKc = S^2 + 2 zeta * wn * s + wn^2
        ii. 0.419Kc = 2 * zeta * wn = 2 * 0.591 * 1.692 = 2
        iii. Kc = 2/.419 = 4.77
        iv. zKc = wn^2 = 1.692^2 = 2.86
        v. z = 2.86/4.77 = 0.6
    c. Results
        i. Kc = 2/.419 = 4.77
        ii. z = 2.86/4.77 = 0.6